

Public Policy and Private-Sector Prosocial Motives: The Case of Greenhouse Gas Emissions

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Motivation

- Both public and private sectors contribute to the common good
 - Combating climate change:
 - Public: Paris Agreement, state-level emissions reduction targets, Inflation Reduction Act of 2022, etc.
 - Private: Shareholder engagement on climate issues
Larry Fink, CEO of BlackRock: *"No issue ranks higher than climate change on our clients' lists of priorities. They ask us about it nearly every day."*

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- Public and private efforts toward similar prosocial goals may interact

Research Question

- **How do public policies affect private-sector prosocial motives?**
 - Unclear ex ante
 - Depends on effects on private sector's extrinsic, reputational, and altruistic motives
 - Overall impact could be either crowding-in or crowding-out

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 - Shared goal between public and private sectors
 - Measurable, interchangeable efforts
 - Partial policy adoption within the sample

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 - Partial policy adoption within the sample
- This study: Greenhouse gas (GHG) emissions reduction
 - 9 U.S. states adopted emissions reduction targets in 2019; 25 states never adopted

Preview of Results (1/2)

- Firms in policy-adopting states experience reduced shareholder pressure post-treatment
 - Lower likelihood of receiving emissions-related proposals (↓ 37.5% of average)
 - Lower support rates (↓ 40.7% of average)

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- No evidence that state-level policies effectively reduce corporate emissions
- Overall, public interventions may unintentionally discourage private-sector prosocial efforts, thereby undermining intended policy effects

Preview of Results (2/2)

- Methodologically, regressions using log-transformed vs. non-transformed individual-level outcome variables $\Rightarrow$ different research questions
 - Different meanings of average treatment effects (ATEs):
 - Log-transformed: Captures average percentage change
 - Non-transformed: Captures average level change
 - Difference between these two ATEs nontrivial, may have opposite signs
 - Especially with dispersed outcomes and heterogeneous treatment effects
- For aggregate outcomes, like climate policy effectiveness, non-transformed outcomes are more appropriate

Literature

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 - Piatti et al. (2023) model taxes & subsidies crowding out private-sector altruistic motives
 - **This paper:** Extends to reputational motives; first empirical study
- Percentage vs. level reduction in emissions
 - Hartzmark and Shue (2023): investors/ESG ratings mistakenly focus on percentage, not level reductions
 - **This paper:** Raises similar concern in academic research using log-transformed emissions
 - Convention in climate finance (Chan and Morrow, 2019; Bartram et al., 2022; Kumar and Purnanandam, 2024; Korganbekova, 2024)
 - Log v.s. non-log discussion broadly relevant for assessing impacts of macro-level factor

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- Since 2006, 26 states adopted emissions reduction targets
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- Example: Colorado's Climate Action Plan to Reduce Pollution (2019)
 - Goals: 26% below 2005 levels by 2025, 50% by 2030, 90% by 2050
 - Specific regulations and programs:
 - Disclosure requirement: Electric utilities must submit plans to reduce emissions 80% by 2030 (SB19-236)
 - Industry standard: Requires local jurisdictions to upgrade energy standards for buildings (HB19-1260)
 - Financial incentives: Grants/low-interest loans for businesses to purchase electric vehicles (SB21-260)

To do: plot the timeline of these targets and Paris Agreement (join, withdraw, rejoin) and the Inflation Reduction Act of 2022

Data and Sample

Data Sources

- Shareholder Proposals: ISS Shareholder Proposals
 - Classified as emission-related if mentioning “GHG”, “climate”, etc.
- Voting Support Rate: ISS Company Vote Results
- Emissions: Greenhouse Gas Reporting Program (GHGRP) by the U.S. (EPA)
 - From 2010, required for facilities emitting $\geq 25,000$ metric tons of CO₂e annually
- Control variables: CRSP/Compustat Merged, CRSP Monthly Stock, Kenneth French's Website, Bureau of Economic Analysis, IBES Detail File with Actuals

Sample Construction:

- Data availability: 2010-2021; event window: $[t-3, t+2]$
 - 11 states (with 15,025 facilities) adopted state-level targets, out of which 9 (with 14,655 facilities) in 2019
 - Focus on these 9 states
- Treatment group: firms/facilities in 9 states adopting targets
- Control group: firms/facilities in 25 states without targets
- Excluded: States adopting targets in other years

Sample Summary

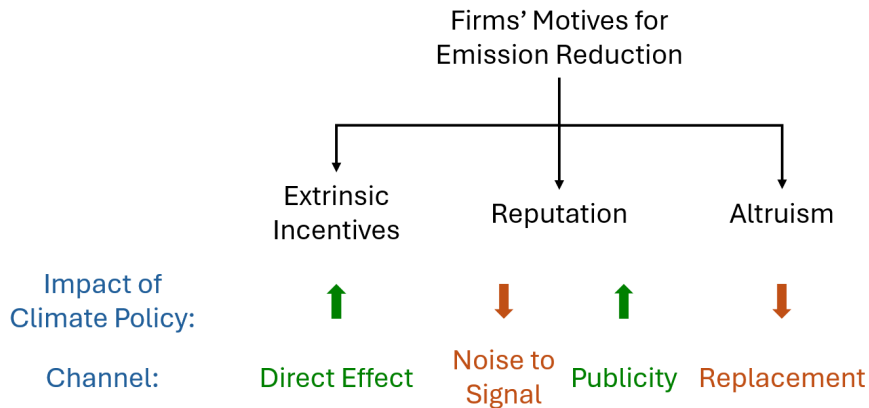
- U.S. panel data, 2016-2021
 - 2,057 firms, 4,763 facilities, 281 emission-related proposals (100 voted on)
- On average,
 - 2.16% of firms receive emission-related proposals each year
 - Firms that ever received emission-related proposals account for 66.4% of emissions
 - 10.8% of proposals are emission-related
 - Average facility emissions: 0.45 million metric tons CO₂e per year

Policy Impact on Private-Sector Prosocial Motives

Theoretical Framework

- Three motives for prosocial behaviour (Bénabou and Tirole, 2006)
- Applied to emissions reduction
 - Extrinsic incentives: Subsidies, taxes, industry standards, etc.
 - Altruism: Sacrifice profits for sustainability
 - Reputation: Project an altruistic image
- Climate policies affect these motives through different channels

Firms' Motives for Emissions Reduction



Empirical Specification

- Panel data regression model with year and firm fixed effects

$$Y_{it} = \beta_0 + \beta_1 Treated_{it} + B_2 X_{it} + d_i + d_t + \varepsilon_{it} \quad (1)$$

with

- Y_{it} : outcome variable of firm i in year t
 - *GHG Proposal Dummy*: indicator of receiving at least one emission-related proposal
 - *GHG Proposal Ratio*: proportion of emission-related shareholder proposals
- $Treated_{it}$: 1 if firm i 's headquartered state has adopted a target by year t , 0 otherwise
- X_{it} : controls
- d_i : firm FE
- d_t : year FE
- ε_{it} : error term, adjusted for heteroskedasticity and state-level clustering

Policy Impact on Emission-Related Proposals (1/2)

Dependent variable =	<i>GHG Proposal Dummy</i>	<i>GHG Proposal Ratio</i>
Treated	-0.0081** (0.00)	-0.0544*** (0.02)
Firm FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
Obs	10274	1140

- After policy adoption
 - Likelihood of receiving emission-related proposals ↓ 0.81% (avg. 2.16%)
 - Proportion of emission-related proposals ↓ 5.44% (avg. 10.77%)

Policy Impact on Emission-Related Proposals (2/2)

- Pre-post comparison within groups
 - Treatment: likelihood ↓ proportion ↓
 - Control: changes much less significant

Policy Impact on Voting Support Rates

- After policy adoption
 - Support rate ↓ 12.78% (avg. 31.38%)

Figures of Policy Impacts on Shareholder Engagement

- Conclusion: State-level targets discourage submission and support for emission-related proposals

Policy Effect on GHG Emissions

Policy Effect on GHG Emissions

- So far: State-level targets crowd out shareholder prosocial motives
- Still unclear: Effectiveness in reducing emissions
 - Depends on whether crowding-out effect fully offsets direct impact
- Apply DID framework to facility-level emissions

Log vs. Non-log Outcome Variables

- Convention: Log-transformed entity-level emissions
 - Captures average percentage change
 - Overlooks emission scale differences across facilities
 - Does not reflect aggregate treatment effect
- My suggestion: Non-transformed entity-level emissions
 - Captures average level change
 - Accounts for scale differences
 - Proportional to aggregate treatment effect
- Climate policy effectiveness should be evaluated using non-transformed emissions

Policy Impact on GHG Emissions

- Using $\ln(Emission)$: Average percentage change $\downarrow$
- Using $Emission$: Average level change $\rightarrow$
 - $\Rightarrow$ No evidence for policy effectiveness in reducing emissions

Figure of Policy Impact on GHG Emissions

Regression Results by Quintiles

- A larger number of lower-emitting facilities $\downarrow$ emissions
- A smaller number of higher-emitting facilities $\uparrow$ emissions
- $\Rightarrow$ Explains seemingly contradictory ATEs between $\ln(\text{Emission})$ and Emission

Policy Impact on State-Level GHG Emissions

- Confirms lack of evidence for policy effectiveness

Paris Agreement

Uncertainty in Policy Effectiveness

- Climate policies not inherently ineffective
 - Depends on balance between direct impact and crowding-out effect
- Effective if:
 - Direct impact $>$ crowding-out effect
 - Publicity enhances reputational motives

Pre-Post Comparisons of the Paris Agreement (1/2)

- After Paris Agreement:
 - Likelihood and proportion of missions-related proposals ↑, support rates ↑

Pre-Post Comparisons of the Paris Agreement (2/2)

- After Paris Agreement: GHG emissions ↓
- Suggestive evidence of crowding-in of private-sector motives and effective emissions reduction

Conclusion

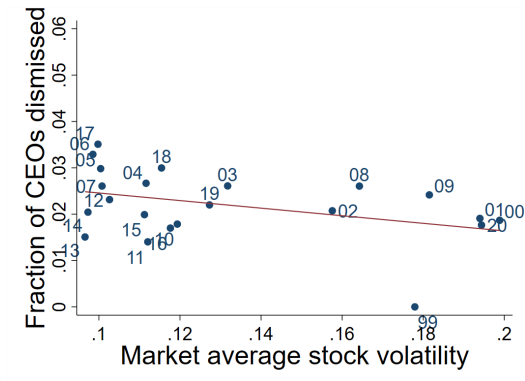
Conclusion

- Empirically examines the impact of public policy on private-sector prosocial motives
 - State-level emissions reduction targets crowd out shareholder motives
 - No evidence of reduced emissions
- Caution for public policy making
 - Public policies are not inherently ineffective or unwarranted
 - Effectiveness may not follow a simple “every little bit helps” logic
- Methodological contribution
 - Log-transformed vs. non-transformed outcome variables have different implications
 - Selection should depend on whether the research question focuses on individual or aggregate outcomes
 - Broad relevance for assessing macro-level factors

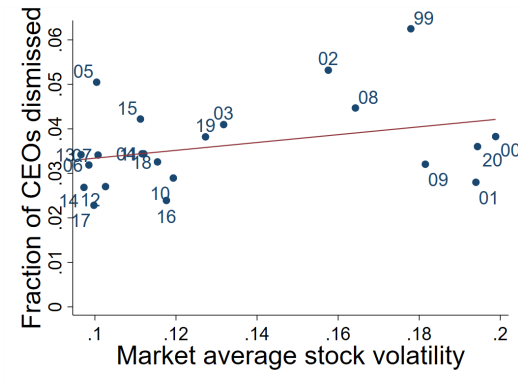
Results

Q1: Are powerful CEOs less likely to be dismissed under higher uncertainty?

CEO Dismissals on the Market Level



Powerful CEOs



Other CEOs

- Uncertainty $\uparrow$: dismissal(powerful CEOs) $\downarrow$; dismissal(other CEOs) $\uparrow$

Empirical Specification

- Panel data regression model with year and industry (or year-by-industry) fixed effects

$$\begin{aligned} \text{Forced turnover}_{it} = & \beta_0 + \beta_1 \text{Uncertainty}_{it} + \beta_2 \text{CEO power}_{it} \\ & + \beta_3 \text{CEO power}_{it} \times \text{Uncertainty}_{it} + B_4 X_{it} \\ & + B_5 X_{it} \times \text{Uncertainty}_{it} + d_I + d_t (\text{or } d_{It}) + \varepsilon_{it} \end{aligned} \quad (2)$$

with

- *Forced turnover*_{it}: forced turnover dummy of firm *i* in year *t*
- *Uncertainty*_{it}: uncertainty of industry *I* in year *t*
- *CEO power*_{it}: the power of the CEO of firm *i* in year *t*
- *X*_{it}: controls
- *d*_I: industry FE; *d*_t: year FE
- *d*_I*t*: industry-by-year FE
- *ε*_{it}: error term, adjusted for heteroskedasticity and industry-level clustering

Regression of CEO Dismissals

Dependent variable =	Forced turnover dummy	
Uncertainty	0.060 (0.14)	
CEO power	0.005 (0.01)	-0.000 (0.00)
CEO power $\times$ Uncertainty	-0.104*** (0.04)	-0.095*** (0.03)
Year FE & Industry FE	Yes	No
Year-Industry FE	No	Yes
Controls	Yes	Yes
Obs	32033	32033

- The dismissal rate of powerful CEOs significantly decreases with uncertainty
- Uncertainty $\uparrow$ one SD $\rightarrow$ dismissal rate(powerful CEO) $\downarrow$ 0.47%
 - Economically significant: the average rate is 2.31%

Q2: Is the fact that powerful CEOs are less likely to be dismissed under higher uncertainty optimal for firms?

Two Theories of CEO Turnover

- Optimal dismissal theory (Gibbons and Murphy, 1990; Bushman et al., 2010)
 - Boards make retention or dismissal decisions in the best interests of shareholders
 - Less dismissals → Firms' preferences for powerful CEOs increase with uncertainty

Two Theories of CEO Turnover

- Optimal dismissal theory (Gibbons and Murphy, 1990; Bushman et al., 2010)
 - Boards make retention or dismissal decisions in the best interests of shareholders
 - Less dismissals → Firms' preferences for powerful CEOs increase with uncertainty
- Entrenchment theory (Shleifer and Vishny, 1989; Hermalin and Weisbach, 1998)
 - Entrenched CEOs can distort the retention or dismissal decisions in their own favour
 - Less dismissals → Powerful CEOs' entrenchment cost increases with uncertainty

Firm Performance

Dependent variable =	Q	
Uncertainty	1.930* (1.12)	3.641** (1.46)
CEO power	0.010 (0.07)	-0.037 (0.06)
CEO power $\times$ Uncertainty	0.412 (0.55)	0.767 (0.49)
Year FE & Firm FE	Yes	Yes
Controls	No	Yes
Obs	28569	28569

- No evidence that powerful CEOs retained in uncertain times underperform
- Similar results if measuring performance by *ROA* or *Sales growth*

Stock return during the 2020 Coronavirus Stock Market Crash

Dependent variable =	Cumulative return Feb 20th to Mar 20th			
Year =	2020		2019	
CEO power	0.030*** (0.01)	0.028*** (0.01)	-0.007 (0.00)	-0.006 (0.00)
Firm Size		0.009** (0.00)		0.001 (0.00)
Constant	-0.411*** (0.01)	-0.417*** (0.05)	-0.015*** (0.00)	-0.016 (0.02)
Controls	No	Yes	No	Yes
Obs	1427	1424	1539	1537

- To mitigate the endogeneity concern of CEO power, use the onset of the COVID-19 pandemic as an unanticipated sudden surge in uncertainty
- Powerful CEOs are favoured by the market in uncertain times

Q3: What mechanisms make powerful CEOs particularly valuable under uncertainty?

Mechanism 1: Better Information Sharing

- In uncertain times, new information is generated at an accelerated rate
- Powerful CEOs may be more willing to share information with the board (Adams and Ferreira, 2007)
- Prediction: powerful CEOs should be especially beneficial for obscure firms

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- In uncertain times, new information is generated at an accelerated rate
- Powerful CEOs may be more willing to share information with the board (Adams and Ferreira, 2007)
- Prediction: powerful CEOs should be especially beneficial for obscure firms
- Method:
 - Divide firm-years into two groups by information asymmetry
 - Run regressions of CEO dismissal within each group

Mechanism 1: Better Information Sharing

	Dependent variable =	Forced turnover dummy	
	Measure for uncertainty =	Stock volatility	Delisting rate
Obscure firms	CEO power $\times$ Uncertainty	-0.120** (0.06)	-0.216*** (0.07)
	Obs	15105	15105
Transparent firms	CEO power $\times$ Uncertainty	-0.046 (0.05)	-0.024 (0.06)
	Obs	16028	16028
	Year FE	Yes	Yes
	Industry FE	Yes	Yes
	Controls	Yes	Yes

- Among obscure firms, powerful CEOs are less likely to be dismissed in uncertain times

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- Among obscure firms, powerful CEOs are less likely to be dismissed in uncertain times
- Insignificant results among transparent firms.

Mechanism 2: Higher Response Speed

- The CEO runs a firm daily, while board directors often have other commitments
- In uncertain times, slower decision-making is more costly
- Prediction: for firms with busier directors, having a CEO with more decision-making power should be especially beneficial

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Busier directors	CEO power $\times$ Uncertainty	-0.141** (0.06)	-0.209** (0.09)
	Obs	15697	15697
Less busy directors	CEO power $\times$ Uncertainty	-0.053 (0.07)	-0.073 (0.07)
	Obs	16336	16336
	Year FE	Yes	Yes
	Industry FE	Yes	Yes
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- Among firms with busier directors, powerful CEOs are significantly less likely to be fired under higher uncertainty

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- Insignificant results among firms with less busy directors

Conclusion

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- Powerful CEOs are less likely to be dismissed under higher uncertainty
- Optimally determined by the board, rather than distorted by CEO entrenchment
- Two potential mechanisms for why powerful CEOs become more preferred under higher uncertainty
 - Better information sharing
 - Higher response speed
- This study challenges the view that CEO power is always manipulative and detrimental